Повтор шагов

(*https://andreyex.ru/linux/kak-otlazhivat-stsenarij-bash/*)

Опция -x для локального скрипта *hello.sh*:

```
admin@somename:~/bin$ bash -x ./hello.sh
++ date '+%Y-%m-%d %H:%M:%S'
+ echo 'Hello, admin. Script launched: 2026-04-03 07:23:28. Number of arguments: 0'
Hello, admin. Script launched: 2026-04-03 07:23:28. Number of arguments: 0
+ '[' 0 -eq 0 ']'
+ echo 'Compiled succesfully'
Compiled succesfully
admin@somename:~/bin$
```

Отладка части локального скрипта *configure_vm.sh*:

```
set_host_variables() {
    set -x

    echo "Enter hostname:"
    read HOSTNAME

    echo "Enter timezone:"
    read TIMEZONE

    set +x

    echo "Enter ip address:"
    read IP_ADDRESS

    echo "Enter ip gateway:"
    read IP_GATEWAY
}
```

```
admin@somename:~/bin$ ./configure_vm.sh host
+ echo 'Enter hostname:'
Enter hostname:
+ read HOSTNAME
some
+ echo 'Enter timezone:'
Enter timezone:
+ read TIMEZONE
Europe/Minsk
+ set +x
Enter ip address:
192.168.100.50
Enter ip gateway:
192.168.100.1
```

Вывод синтаксических ошибок из *configure_vm.sh* и *hello.sh*:

```
admin@somename:~/bin$ bash -n ./configure_vm.sh
admin@somename:~/bin$ bash -n ./hello.sh
admin@somename:~/bin$
```

Вывод команд перед их выполнением:

```
admin@somename:~/bin$ bash -v ./hello.sh
#!/bin/bash

echo "Hello, ${1:-$USER}. Script launched: $(date +"%Y-%m-%d %H:%M:%S"). Number of arguments: "$#""
Hello, admin. Script launched: 2026-04-03 08:01:19. Number of arguments: 0

if [ $? -eq 0 ]; then
        echo "Compiled succesfully";
else
        echo "Compiled with error";
fi
Compiled succesfully
```

Отладка скрипта имитирующего проверку доступности сервисов по урлам

1. Проверка скрипта через *bash -x* показала, что нет нужного для исполнения файла *urls.txt, "[!"* неверный синтаксис, *PIDS* команда не нашлась:

```
admin@some:~/bin/urls_check$ bash -x urls_check.sh
+ INPUT_FILE=:-urls.txt
+ MAX_PARALLEL=2
+ declare -a PIDS
+ '[!' -f INPUT_FILE ']'
urls_check.sh: line 13: [!: command not found
+ count=0
urls_check.sh: line 33: :-urls.txt: No such file or directory
++ 'PIDS[@]'
urls_check.sh: line 34: PIDS[@]: command not found
+ wait
+ echo Готово.
Готово.
```

2. Вывод *shellcheck* следующий (ругается на if выражения):

```
admin@some:~/bin/urls_check$ shellcheck urls_check.sh

In urls_check.sh line 13:
if [! -f INPUT_FILE ]; then
   ^-- SC1035 (error): You need a space after the [ and before the ].


In urls_check.sh line 28:
if [ $count-ge $MAX_PARALLEL ]; then
^-- SC1009 (info): The mentioned syntax error was in this if expression.
   ^-- SC1073 (error): Couldn't parse this test expression. Fix to allow more checks.
           ^-^ SC1108 (error): You need a space before and after the -ge .
               ^-- SC1072 (error): Expected test to end here (don't wrap commands in []/[[]]). Fix any mentioned problem
s and try again.

For more information:
  https://www.shellcheck.net/wiki/SC1035 -- You need a space after the [ and ...
  https://www.shellcheck.net/wiki/SC1108 -- You need a space before and after...
  https://www.shellcheck.net/wiki/SC1072 -- Expected test to end here (don't ...
```

3. После добавления *urls.txt*, исправления объявления переменной *INPUT_FILE* и использования её в блоке *if bash -x* выводит следующее (+ разделил *ge* от вывода):

```
admin@some:~/bin/urls_check$ bash -x urls_check.sh
+ INPUT_FILE=urls.txt
+ MAX_PARALLEL=2
+ declare -a PIDS
+ '[' '!' -f urls.txt ']'
+ count=0
+ IFS=
+ read -r line
++ echo '26*'
+ line='26*'
+ line='26*'
+ line='26*)'
+ '[' -z '26*)' ']'
+ PIDS+=0
+ process_url '26*)'
++ count+1
+ local 'url=26*)'
+ echo 'проверка: 26*)'
проверка: 26*)
+ '[' -n '' ']'
+ sleep 1
urls_check.sh: line 34: count+1: command not found
+ count=0
+ '[' 0 -ge 2 ']'
+ IFS=
+ read -r line
++ 'PIDS[@]'
urls_check.sh: line 41: PIDS[@]: command not found
+ wait
+ echo 'стратус: OK'
стратус: OK
```

А *shellcheck*:

```
admin@some:~/bin/urls_check$ shellcheck urls_check.sh

In urls_check.sh line 32:
process_url $line &
            ^---^ SC2086 (info): Double quote to prevent globbing and word splitting.

Did you mean:
process_url "$line" &


In urls_check.sh line 33:
PIDS+=$(($1))
^--^ SC2179 (warning): Use array+=("item") to append items to an array.


In urls_check.sh line 36:
wait $(PIDS[0])
     ^-------^ SC2046 (warning): Quote this to prevent word splitting.
       ^-----^ SC2211 (warning): This is a glob used as a command name. Was it supposed to be in ${..}, array, or is it missing quoting?


In urls_check.sh line 37:
PIDS="-${PIDS[@]:1}"
^--^ SC2178 (warning): Variable was used as an array but is now assigned a string.
     ^------------^ SC2124 (warning): Assigning an array to a string! Assign as array, or use * instead of @ to concatenate.


In urls_check.sh line 40:
done < $INPUT_FILE
       ^---------^ SC2086 (info): Double quote to prevent globbing and word splitting.

Did you mean:
done < "$INPUT_FILE"


In urls_check.sh line 41:
wait $(PIDS[@])
     ^-------^ SC2046 (warning): Quote this to prevent word splitting.
       ^-----^ SC2211 (warning): This is a glob used as a command name. Was it supposed to be in ${..}, array, or is it missing quoting?

For more information:
  https://www.shellcheck.net/wiki/SC2046 -- Quote this to prevent word splitt...
  https://www.shellcheck.net/wiki/SC2124 -- Assigning an array to a string! A...
  https://www.shellcheck.net/wiki/SC2178 -- Variable was used as an array but...
admin@some:~/bin/urls_check$
```

После добавления в места вставки переменных кавычек и фикса опечатки *“INPUT FIL”E* стало выглядеть следующим образом:

```
admin@some:~/bin/urls_check$ bash -x urls_check.sh
+ INPUT_FILE=urls.txt
+ MAX_PARALLEL=2
+ declare -a PIDS
+ '[' '!' -f urls.txt ']'
+ count=0
+ IFS=
+ read -r line
++ echo '26*'
+ line='26*'
+ line='26*'
+ line='26*)'
+ '[' -z '26*)' ']'
+ PIDS+=0
+ process_url '26*)'
+ local 'url=26*)'
+ echo 'проверка: 26*)'
проверка: 26*)
+ '[' -n '' ']'
+ sleep 1
++ count+1
urls_check.sh: line 34: count+1: command not found
+ count=0
+ '[' 0 -ge 2 ']'
+ IFS=
+ read -r line
++ 'PIDS[@]'
urls_check.sh: line 42: PIDS[@]: command not found
+ wait
+ echo 'стратус: OK'
стратус: OK
+ echo Готово.
Готово.
```

Убрал двойные скобки где не нужно:

```
admin@some:~/bin/urls_check$ bash -x urls_check.sh
+ INPUT_FILE=urls.txt
+ MAX_PARALLEL=2
+ declare -a PIDS
+ '[' '!' -f urls.txt ']'
+ count=0
+ IFS=
+ read -r line
++ echo '26*'
+ line='26*'
+ line='26*'
+ line='26*)'
+ '[' -z '26*)' ']'
+ PIDS+=($!)
+ count=1
+ '[' 1 -ge 2 ']'
+ IFS=
+ read -r line
+ wait 4593
+ process_url '26*)'
+ local 'url=26*)'
+ echo 'проверка: 26*)'
проверка: 26*)
+ '[' -n '' ']'
+ sleep 1
+ echo 'стратус: OK'
стратус: OK
+ echo Готово.
Готово.
admin@some:~/bin/urls_check$
```

Исправил *$(PIDS[0])* на *“${PIDS[0]}”* и *“-${PIDS[@]:1}”* на *(“${PIDS[@]:1}”)*

```
admin@some:~/bin/urls_check$ shellcheck urls_check.sh

In urls_check.sh line 36:
                wait $(PIDS[0])
                     ^--------^ SC2046 (warning): Quote this to prevent word splitting.
                       ^-----^ SC2211 (warning): This is a glob used as a command name. Was it supposed to be in ${..},
array, or is it missing quoting?


In urls_check.sh line 37:
                PIDS="-${PIDS[@]:1}"
                ^--^ SC2178 (warning): Variable was used as an array but is now assigned a string.
                     ^-------------^ SC2124 (warning): Assigning an array to a string! Assign as array, or use * instead
 of @ to concatenate.

For more information:
  https://www.shellcheck.net/wiki/SC2046 -- Quote this to prevent word splitt...
  https://www.shellcheck.net/wiki/SC2124 -- Assigning an array to a string! A...
  https://www.shellcheck.net/wiki/SC2178 -- Variable was used as an array but...
admin@some:~/bin/urls_check$ nano urls_check.sh
admin@some:~/bin/urls_check$ shellcheck urls_check.sh
```

Также плохо выглядел момент, где происходил вывод урлов и в конце к каждому добавлялось **)*. Когда убрал обработку урлов в коде, а оставил обычный вывод получилось вот так:

```
admin@some:~/bin/urls_check$ ./urls_check.sh
проверка: https://www.wikipedia.org/
проверка: https://localhost:5980/
стратус: OK
стратус: OK
Готово.
admin@some:~/bin/urls_check$
```